JNTUH USED PAPERS NOV 20

Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, November/December - 2020
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

1.a) List and explain various challenges that are occurred in the process of designing distributed systems.
b) What are the specific functional properties of a video streaming network system to satisfy the expected QoS? [7+8]

2.a) Explain about Christian's method for synchronizing clocks.
b) Write about *Maekawa's* voting algorithm. [8+7]

3.a) What is the need for Serialization? Design a set of constructors and its corresponding method for a serialized object "Employee" using Java class using CORBA IDL.
b) How are unused remote objects handled by Java garbage handling? Explain how the algorithm helps to avoid failure. [8+7]

4.a) What are the characteristics of file System? Explain briefly.
b) Compare the relative advantages of X.500 with GNS. [7+8]

5.a) Explain how the time-stamp approach helps in overcoming the lost-update problem. Prepare a modified algorithm and explain how the inconsistencies are removed.
b) Compare the operations of Two-phase commit protocol over three-phase commit protocol. Identify the advantages of each method. [8+7]

6.a) Discuss how Denial of service attack takes place in a distributed network system.
b) Explain about Software and Hardware service layers in distributed system. [8+7]

7.a) Explain how Berkley's algorithm tries to resolve the pitfalls of Christian's algorithm.
b) What is the use of bully algorithm? Illustrate with an example. [8+7]

8.a) List and explain various design issues of RMI.
b) Explain in detail about Sun RPC. [8+7]

---ooOoo---